Handoff 13: Public website and product design direction (SITE-01)

Date: 2026-10-26 (America/Toronto)
Written by: Claude Code (implementer), overnight session
For: the human maintainer, and ChatGPT or Claude chat (reviewer) without repository access
Status: **in_review**: one pull request, not merged, not deployed. **Documentation only**: no application code, content, tests, settings or deployment changed.

> **How to use this file:** upload it with `build.md` and `docs/plan/site-design.md`. Handoff 08 (CSP nonces) helps with section 4 of the design. The pull request diff is the source of truth (`build.md` section 14).

Starter prompt for the chat assistant

```
You are reviewing Python Academy SITE-01, a design direction (documentation
only) for a public, scroll-driven story website that ends in a button into
the product, and for calm polish of the product itself. I'm attaching
build.md, docs/plan/site-design.md and this handoff. Compare the three story
concepts for a grade 9 beginner, check that the motion and performance
rules are realistic for a school laptop, that the animation approach really
works with a nonce-based Content-Security-Policy that blocks inline style
attributes, and that the architecture recommendation (a separate static
site) is sound on cost, security and maintenance. Explain at a first-year
computer engineering level.
```

1. Where we are

Stage	What	Status
11	ENG-03 browser behaviour checks	PR #17, open
14	DOC-01, DOC-02 decision log	PR #18, open
13	SITE-01 site and product design direction	**in_review**: this PR. It does not depend on #17 or #18, and merges cleanly with both (section 6)

2. Handoff record (build.md template)

Field	Value
Task	SITE-01: design direction for a public scroll-story landing site and for product polish: concepts, illustrations, motion, technology, architecture, accessibility, staged plan, backlog rows, questions
Human goal	A memorable public face that explains the mission and leads into a calm, fast product, without hurting speed, security, accessibility or cost
Relevant build.md sections	1 (mission), 3 (learner modes), 4 (teaching contract: calm, one idea at a time), 16 (brand is an open decision)
Repository and branch	`torontojohn/python-academy`, branch `docs/site-design`
Base commit	`27ab31b` (`main` after PR #16)
Implementer / reviewer	Claude Code / owner, plus ChatGPT or Claude chat
Deployment status	Not applicable (documentation only); not merged
New dependencies	None

Files changed

File	Change
`docs/plan/site-design.md`	New. Sections 0 to 10, as the task listed: today's look; three concepts; illustrations; motion and the performance budget; technical approach; architecture, domain and rename; accessibility; product polish; staged plan; backlog rows; questions
`build.md`	Section 2 lists the new document

Summary of the recommendations

- **Concept:** B, "Your Code Builds the World", with a small map of the coming courses in its mission scene (the owner chooses).
- **Illustrations:** hand-written SVG in the style of the existing castle scene.
- **Motion:** CSS first, with no scroll-jacking and no loading screen that hides content; a short intro, a pause button, and full support for reduced motion.
- **Architecture:** a separate static site in the same repository, with no login and no cookies. It is free on Render (to confirm), served from a CDN, and needs no changes to the app.
- **Product (SITE-03):** polish, not a redesign: named colours, catalog grouping, course emblems, and a short celebration on passing.

3. Decisions and why

The design document makes **proposals**; the owner makes the decisions (section 10 of the design). The choices made in writing it:

Decision	Options considered	Reason
Suggest concept B (code and picture side by side) as the default	A: Lantern Path; C: The Map	B shows the real product honestly (it *is* unit 1), teaches while it markets (predict first), and needs the least art. C's map is folded into B's mission scene to show growth beyond Python
Recommend hand-written SVG illustrations	Commissioned art; AI-generated art	Free; matches the art already in the product; tiny files; each part can move; reviewable as code; clear ownership. AI images are hard to keep consistent, cannot be animated in parts, and have unsettled rights; commissioned art costs money (free-first rule)
CSS scroll-driven animations first, with no library	Motion, GSAP, Lottie, smooth-scroll libraries, 3D	Nothing to download, nothing to break the policy. Lottie's full player has used `eval`; smooth scrolling takes over the wheel; 3D is too heavy for school laptops
A rule: no `style` attributes in server-rendered HTML	Allowing them	Our page policy is `style-src 'self'`, which blocks inline style attributes, and a nonce cannot allow them. Some React animation helpers write starting positions that way
Recommend a separate static site	Story inside the app at `/`	No extra cost; landing traffic cannot slow learners; nothing to attack (no cookies or API); fastest first load; **no changes** to sign-in redirects or invitation links. Cost: a second small deploy
A short intro instead of a loading screen	A full loading screen, as on the inspiration site	A screen that hides content hurts speed and accessibility. The intro plays over content that is already there and is skipped under reduced motion (question 6)
Budgets taken from Google's "good" Core Web Vitals (LCP 2.5 s, INP 200 ms, CLS 0.1), on a throttled profile	No budget	A school Chromebook is slow, so a concrete target lets each later PR be measured
Primary call to action: "Try the first lesson, no account needed" (`/preview?lesson=hello`)	"Sign up" first	It already exists, saves nothing, and lets a beginner feel success before any account
Do not copy hackcanada.org	—	The owner's instruction; the concepts are built only from our unit 1 world

Decision	Options considered	Reason
Do not edit `docs/plan/backlog.md`; put the `build.md` row in a different place from PR #18's	Editing both	Avoids merge conflicts with open PRs (checked, section 6)

4. Bugs found and how they were fixed

None: no code changed. Two findings recorded in the design, for later stages:

- **About 70 hard-coded colours** in `frontend/app/style.css`, many of them near-duplicate greens. Not a bug, but it makes consistent polish hard. SITE-03 stage 5 replaces them with about a dozen named colours.
- **A trap for later:** server-rendered `style="…"` attributes would be reported by today's Report-Only page policy and blocked once it is enforced. Recorded as a rule in design section 4.1, so the story and SITE-03 do not run into it.

5. How it was tested

- **What "tested" means for a design document:** it is checked for accuracy against the code and earlier evidence, not run.
- Colours, the font, the existing SVG scene and the reduced-motion rules were read from `frontend/app/style.css` and `adventure.tsx` at `27ab31b`, by a script that listed every colour, font, media query and animation.
- The look was checked against the handoff 03 and 04 screenshots.
- The cost and rendering facts come from handoff 08's measurements.
- Claims about browser support, Render's free static sites and custom domains, and library licences are marked "confirm at decision time": they were not re-checked online in this session.
- **Link check:** every relative link in `site-design.md` resolves.
- **Merge check:** `git merge-tree` shows no conflicts with PR #17 (`eng/browser-checks`) or PR #18 (`docs/decision-log`).
- **Full test suites,** run although nothing executable changed:

Command	Result
`cd backend && .venv/bin/python -m pytest -q`	`213 passed, 9 skipped`
`cd frontend && npm run build`	Compiled successfully
`cd frontend && CI=1 npm run test:e2e`	`30 passed`

- **Mocks:** none; documentation only.

- **Deliberate-break controls:** not applicable; no code. Each build stage in design section 8 comes with its own tests and live checks.

6. Backlog rows to add (not edited, as instructed)

From design section 9, in the `build.md` format:

ID	Priority	Item	Depends on	Status
SITE-01	Next	Design direction for the public story site and the product's look (`docs/plan/site-design.md`): three story concepts, illustration approach, motion and performance budget, architecture, accessibility, staged plan	None	in_review (handoff 13)
SITE-02	Next	Public landing story site: the chosen concept as a separate static site (proposed), built in three PRs: skeleton with still scenes and strict headers; scroll motion with reduced motion and a pause button; interactive moments. Within the section 3 budget, 0 axe violations	SITE-01 decisions (concept, architecture)	proposed
SITE-03	Later	Product polish to match the story, not a redesign: colour and spacing tokens, catalog grouping with course emblems, course header art, a short lesson celebration; calm and fast	SITE-02 (shared style)	proposed
SITE-04	Later	Custom domain and possible rename: choose a name, buy a domain (paid, owner decision), DNS, `APP_ORIGIN` and `API_ORIGIN`, Google OAuth settings, redirects, one-time sign-in again	Owner decision on name and cost; `build.md` section 16 (brand)	proposed

Also add a stage row: "13. SITE-01 | Site and product design direction | in_review (handoff 13)".

7. Assumptions

1. **Handoff number 13,** as asked. PR #18 used 14, so the files sort in that order.
2. **Grade 9 first:** the concepts are written for a grade 9 beginner, with other students and hackathon teams second, as the task said.
3. **"Wow" without risk:** taken to mean striking illustration and smooth, meaningful motion, not heavy effects. The performance and accessibility budgets are firm limits.
4. **Free first:** paid items (an illustrator, a domain) are offered as options with the owner deciding, never as requirements.

8. Live checks for you

Nothing is deployed by this PR. Instead:

- [ ] **Read the three concepts** in `docs/plan/site-design.md` section 1, and the comparison table. Pick one, or a mix.
- [ ] **Look at today's scene** on `https://python-academy-dji8.onrender.com/preview?lesson=hello`, the castle picture. The recommended illustration style (section 2) extends exactly this.
- [ ] **Try reduced motion today:** turn on "Reduce motion" (macOS: System Settings → Accessibility → Display) and reload a lesson. The stars and moon stop moving. The story site would behave the same way, with a still version of every scene.
- [ ] **See the catalog's busyness:** open `/courses`. Ten "Coming soon" cards fill most of the page; design section 7 proposes a quieter "Coming later" group.
- [ ] **Answer section 9** below, so SITE-02 can start.

9. Questions for the owner

From design section 10:

1. **Concept:** A, B, C, or B with a small map (the suggested default)?
2. **Architecture:** a separate static site (recommended), or inside the app with the product moved to `/app`?
3. **Illustrations:** hand-written SVG (recommended, free), or quotes from an illustrator for one hero image?
4. **Name and domain:** keep "Python Academy" for now, or choose a broader name first? Willing to pay for a domain, and when?
5. **Launch timing:** soon, or after unit 2 exists?
6. **Intro:** is a short intro over visible content close enough to the loading screen you liked?
7. **Teams and teachers:** a section on the landing page, or a separate page later?

10. Next concrete action

The owner reviews this PR and answers section 9. Then SITE-02 stage 2 (the static story skeleton with still scenes and strict headers) is one small PR. The owner creates the new Render static site from `render.yaml` when it is merged.